Translating Maps Between the Senses: Multimodal AI for Spatial Understanding

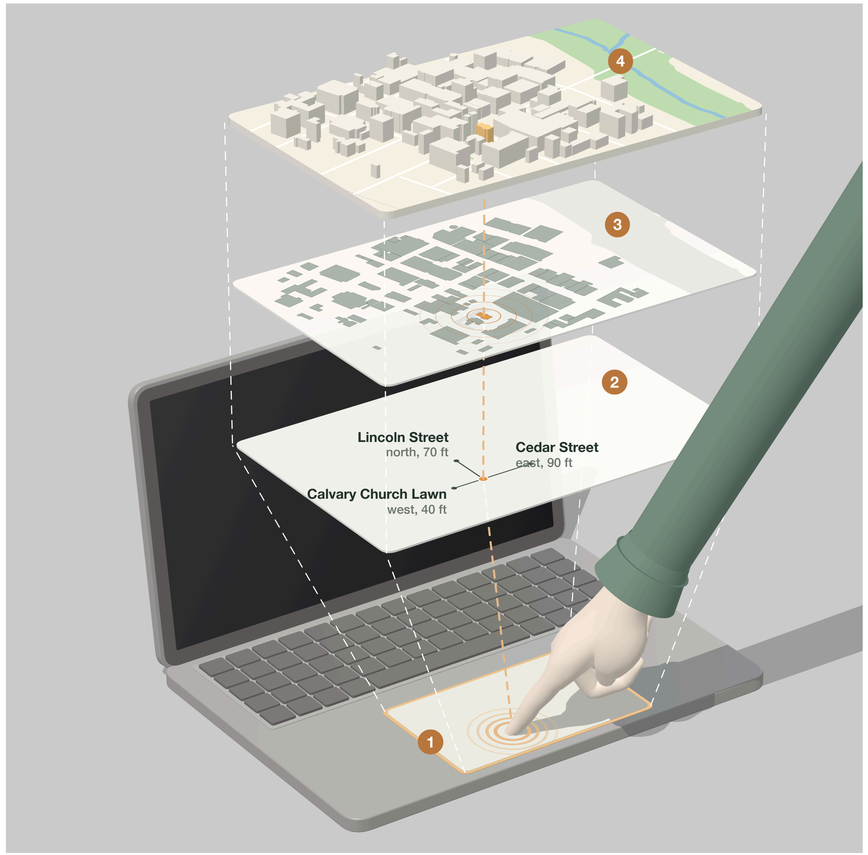
Li Liu[1], Yihe Wang[1], Jiaming Qu[2], Ashmita Dua[1], David T. Lee[1], Leilani H. Gilpin[1]
[1]University of California, Santa Cruz [2]Amazon Corresponding author: lgilpin@ucsc.edu
[2]Work unrelated to position at Amazon

UCSC AIEA Lab

UC SANTA CRUZ

Project page

TouchingSpace An audio-haptic map assistant, co-designed with a blind user [1, 5, 6]



Legend

1 Trackpad	2 Finger context	3 Haptic regions	4 Real map
Standard laptop trackpad, nothing to attach	Direction and distance to nearby places	A hum grows near a place, vibration marks its edge	Streets and buildings from OpenStreetMap

Q&A example

Reader **"What is this place?"**

Model "You are inside Parish Hall, a medium-sized rectangular building in the south of the map."

Participant quotes

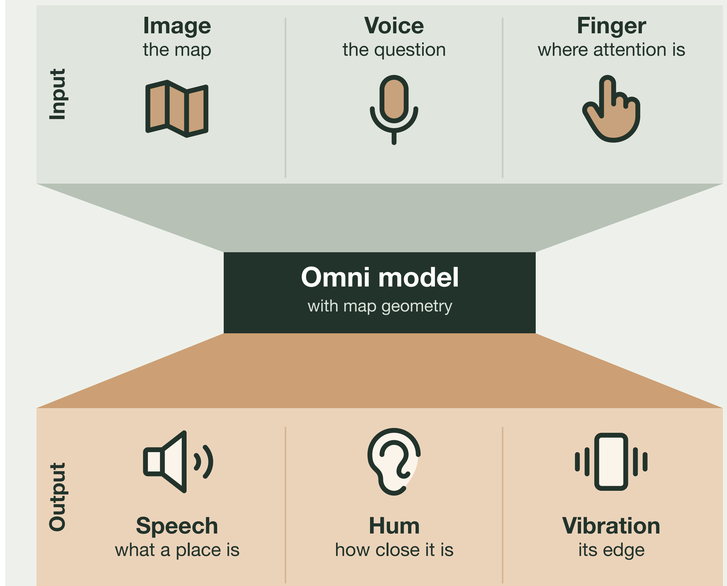
"I didn't know how to read a map, but now I can."

"No one tells you. [The mental map] could be that way forever."

"I can tell someone sighted how to get somewhere now."

"I thought it would be hard for me to build a mental map, but I was able to kind of draw that L-shape or square shape of the streets."

Multimodal input and multisensory output



Research question



How can AI use different senses to convey information and enhance human perception?

Sound, touch, and speech each reveal a different side of space, and each can check the others [2, 3].

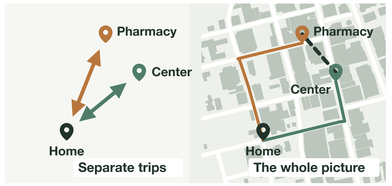
Goal. Real-time answers about where you point, on one GPU.

User feedback from 14 blind and low-vision readers



To know what to search

A search box only answers what you think to ask. Under the finger, readers met places they did not know to ask about.



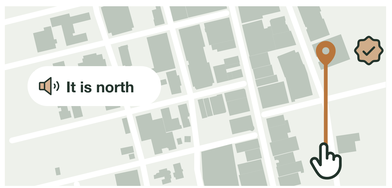
To make a plan

Step-by-step answers keep trips apart. The whole picture joins them [4]: one reader's pharmacy was five minutes from the center.



To build a picture

"I started getting a picture in my mind of where everything was."



To check what the AI says

Readers cross-checked one channel against the others, and against what they already knew.

Broader applications

Robotics	Link speech to a pointing hand
Spatial AI	Test reasoning between places
Digital twins	Explore plans and sites by touch
Wearables	Use gaze or touch as context
Recommendation	Proactively suggest places

Future directions

Proactive adaptation. Infer what a person has learned from video and touch, and adapt the map to gaps and errors they cannot see.

New senses. Translate sensor signals, such as infrared or magnetic fields, into touch.

Two-way learning. Let people correct the model through touch while the model informs them.

Measuring information transfer. Find which modalities work best together, and how much information survives each conversion.

Implementation and results

Voice	Interchangeable commercial or open models
Model	Qwen3-Omni-30B-A3B-Instruct [7]
Device	1 NVIDIA A100 80 GB, holding all three model stages
Software	CUDA, CUDA Graphs, FlashAttention, vLLM-Omni
Response	1.53 s median to first sound
Load	8 users at once
Benchmark	Spoken questions on 3 real maps, scored against map geometry

Question category	n	Gemini 2.5 Flash Native Audio [8]	Qwen3-Omni [7]
Direction and distance from the finger	89	98%	97%
Direction and distance between two places	56	43%	57%
Nearby places	52	73%	75%
Identity of a place	39	87%	85%
Attributes of a place	43	88%	93%
Region of the map	57	86%	77%
Whether a place exists	57	98%	98%
All	**393**	**83.0%**	**84.0%**

Both models find relations between two places hardest.

1. L. Liu, Y. Wang, J. Qu, A. Dua, D. T. Lee, L. H. Gilpin, "'I didn't know how to read a map, but now I can': TouchingSpace, an Audio-Haptic Map for Blind and Low-Vision Readers," arXiv:2609.36404, 2026.
2. P. Bach-y-Rita et al., "Vision Substitution by Tactile Image Projection," Nature, 1969.
3. M. O. Ernst, M. S. Banks, "Humans Integrate Visual and Haptic Information in a Statistically Optimal Fashion," Nature, 2002.
4. J. H. Larkin, H. A. Simon, "Why a Diagram is (Sometimes) Worth Ten Thousand Words," Cognitive Science, 1987.
5. J. E. Froehlich et al., "StreetReaderAI: Making Street View Accessible Using Context-Aware Multimodal AI," UIST, 2025.
6. L. Cavazos Quero et al., "Jido: A Conversational Tactile Map for Blind People," ASSETS, 2019.
7. J. Xu et al., "Qwen3-Omni Technical Report," arXiv:2509.17765, 2025.
8. Gemini Team, Google, "Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities," arXiv:2507.06261, 2025.